18PTNMB4

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Problem

Given a real number a, we define a sequence by $x_0 = 1$, $x_1 = x_2 = a$, and $x_{n+1} = 2x_n x_{n-1} - x_{n-2}$ for $n \geq 2$. Prove that if $x_n = 0$ for some n, then the sequence is periodic.

Solution. Say $|a| > 1$. We will show using induction that $|x_n| > |x_{n-1}|$ for all $n \geq 1$.
For $n = 1, 2$, we have $|x_1| = |x_2| = |a| > 1$.
Assume that $|x_k| > |x_{k-1}|$ for all $k \leq n$. Then, $|x_{n+1}| = |2x_n x_{n-1} - x_{n-2}| \geq 2|x_n||x_{n-1}| - |x_{n-2}| > |x_n|(2|x_{n-1}| - |x_{n-2}|) > |x_n|$.

If $|a| < 1$ then let $a = \cos\theta$ for some $\theta \in (0, \pi)$. Let $F_0, F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ be the Fibonacci sequence. We will show using induction that $x_n = \cos(F_n\theta)$.
For $n = 0, 1, 2$, we have $x_0 = 1 = \cos(0\theta)$, $x_1 = a = \cos(\theta) = \cos(F_1\theta)$, and $x_2 = a = \cos(\theta) = \cos(F_2\theta)$.
Assume that $x_k = \cos(F_k\theta)$ for all $k \leq n$. Then, $x_{n+1} = 2x_n x_{n-1} - x_{n-2} = 2\cos(F_n\theta)\cos(F_{n-1}\theta) - \cos(F_{n-2}\theta)$. Since we have $\cos(F_{n+1}\theta) + \cos(F_{n-2}\theta) = 2\cos(F_n\theta)\cos(F_{n-1}\theta)$, we have $x_{n+1} = \cos(F_{n+1}\theta)$.

If $x_n = 0$ for some n, then $\cos(F_n\theta) = 0 \implies F_n\theta = \dfrac{\pi}{2} + k\pi$ for some integer k. Since θ is a rational multiple of π, let $\theta = \dfrac{p}{q}\pi$ for some positive integers p and q. Then, $F_n\dfrac{p}{q}\pi = \dfrac{\pi}{2} + k\pi \implies F_n\dfrac{p}{q} = \dfrac{1}{2} + k$.
Observe that, if F_n work then $F_n + 2mq$ works as well for any integer m. Since F_n is periodic $\pmod{2q}$, we are done!